A counterexample without quasi-excellence

Abstract

We record a simple affine counterexample showing that the quasi-excellence hypothesis cannot be omitted from the strong-generation theorem for $D^b_{\mathrm{coh}}(X)$. The example is noetherian, one-dimensional, affine, separated, and hence quasiseparated; the only missing hypothesis is quasi-excellence.

The example

For a noetherian ring A, write $\operatorname{Reg}(A) \subseteq \operatorname{Spec} A$ for the regular locus. Recall that A is J-0 if $\operatorname{Reg}(A)$ contains a nonempty open subset, J-1 if $\operatorname{Reg}(A)$ is open, and J-2 if every finite type A-algebra is J-1. A quasi-excellent ring is, in particular, J-2.

Let R be Hochster's one-dimensional noetherian domain which is not J-0 [1, Example 1]. Equivalently, $\operatorname{Reg}(R)$ contains no nonempty open subset of $\operatorname{Spec} R$. Set

$$X = \operatorname{Spec} R.$$

Proposition 1. *The scheme X is noetherian, quasicompact, separated, quasiseparated, and one-dimensional, but it is not quasi-excellent. Moreover $D^b_{\mathrm{coh}}(X)$ has no strong generator.*

Proof. Since X is affine and R is noetherian of dimension one, X is noetherian, quasicompact, separated, quasiseparated, and one-dimensional.

We first show that X is not quasi-excellent. If R were quasi-excellent, then R would be J-2. Applying the definition of J-2 to the finite type R-algebra R itself, R would be J-1, so $\operatorname{Reg}(R)$ would be open. Since R is a domain, the generic point (0) belongs to $\operatorname{Reg}(R)$, because $R_{(0)}$ is the fraction field of R. Thus $\operatorname{Reg}(R)$ would contain a nonempty open subset, i.e. R would be J-0. This contradicts the choice of R.

It remains to prove non-generation. Since X is affine noetherian, there is an equivalence

$$D^b_{\mathrm{coh}}(X) \simeq D^b(\operatorname{mod} R).$$

Suppose, for contradiction, that $D^b_{\mathrm{coh}}(X)$ had a strong generator G. Under the above equivalence, G gives a strong generator of $D^b(\operatorname{mod} R)$; in particular, it is a classical generator.

Consider the singularity category

$$D_{\mathrm{sg}}(R) := D^b(\operatorname{mod} R)/\operatorname{Perf}(R).$$

The Verdier quotient functor

$$q\colon D^b(\operatorname{mod} R) \longrightarrow D_{\mathrm{sg}}(R)$$

is triangulated, so it sends the thick subcategory generated by G onto a thick subcategory generated by qG. Since G classically generates $D^b(\operatorname{mod} R)$, the object qG classically generates $D_{\mathrm{sg}}(R)$.

By Iyengar–Takahashi [2, Lemma 2.9], if the singularity category $D_{\mathrm{sg}}(R)$ has a classical generator, then R is J-1. Since R is a domain, J-1 implies J-0: indeed $(0) \in \operatorname{Reg}(R)$, so an open regular locus is a nonempty open regular locus. This again contradicts Hochster's choice of R. Therefore $D^b_{\mathrm{coh}}(X)$ has no strong generator. In fact, the same argument shows that it has no classical generator. □

Remark 1. *Thus the quasi-excellence hypothesis is genuinely doing work: the counterexample keeps noetherianity, finite dimension, separatedness, and quasiseparatedness. The failure is already affine.*

References

[1] M. Hochster, *Non-openness of loci in Noetherian rings*, Duke Math. J. **40** (1973), 215–219.

[2] S. B. Iyengar and R. Takahashi, *Openness of the regular locus and generators for module categories*, Acta Math. Vietnam. **44** (2019), no. 1, 207–212.